Module - 3

Adaptation of Anatomical Principles for Bioengineering Design

Topics

- Brain as a CPU System
- Eye as a Camera System
- Heart as a Pump System
- Lungs as a Purification System
- Kidney as a Filteration System

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Module - 3 Question Bank

1. Explain Brain as a CPU System - 10M

(or)

Explain the Architecture of Brain - 10M

2. Explain the Architecture of Eye → 10M

(or)

Explain Eye as a Camera System → 10M

3. Explain the Architecture of Heart → 10M

(or)

Explain Heart as a Pump System → 10M

4. Explain the Architecture of lungs → 10M

(or)

Explain Lungs as Purification System. → 10M

5. Explain the Architecture of kidney ? → 10M

(or)

Kidney as a Filteration System ? → 10M

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Brain is the centre of the Nervous System in Humans. It is located With in Skull. The Brain is Composed of Billions of specialised cells called Neurons. These Neurons form Network and pathways throughout the Brain and rest of the Body.

Architecture of Brain

(*) Brain is a Complex Organ it consists several Distincts regions each With its Own specialised functions.

Structure of Brain Consists

(1) Cerebrum - is the Largest part of the Brain. It is divided into two Hemisphere [Right & Left]. These two are Connected by bundle of Nerve fibres.

Cerebrum is divided into 4 Lobes

(1) Frontal Lobe → Reasoning thought
(2) Parietal Lobe → Integrates sensory information
(3) Temporal Lobe → Process auditory information from ears.
(4) Occipital Lobe → Processing Visual information from the eyes.

(2). Cerebellum - is located at the Back of the Brain, below the Cerebrum.

* Recives Sensory information from Muscles and joints, Helps in Movements

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③ Diencephalon - is found between Cerebrum and Midbrain. it contain two structures Thalamus & Hypothalamus.

ⓐ Thalamus - Recives Sensory Information from Various Organs and directs it to Appropriate regions of the cerebral Cortex for processing

ⓑ Hypothalamus - Controles and regulates temperature, Water Balance, sleep etc.

④ Midbrain - Located below cerebrum
* Responsible for Eye and Auditory Reflexes.

⑤ Pons - Located below Midbrain [In Brainstem]
* Responsible for certain reflex actions.
Functions → chewing, Tasting, Saliva production.

⑥ Medulla Oblongata → Located at the Bottom of the Brain stem
* Connected to the spinal cord.
* Regulates Heart & Blood Vessel function i.e Digestion, Respiration, Blood pressure

⑦ Brainstem - Brainstem is located at the Base of the Brain. Helps in Regulating Basic involuntary actions like Heart rate, Breathing etc.

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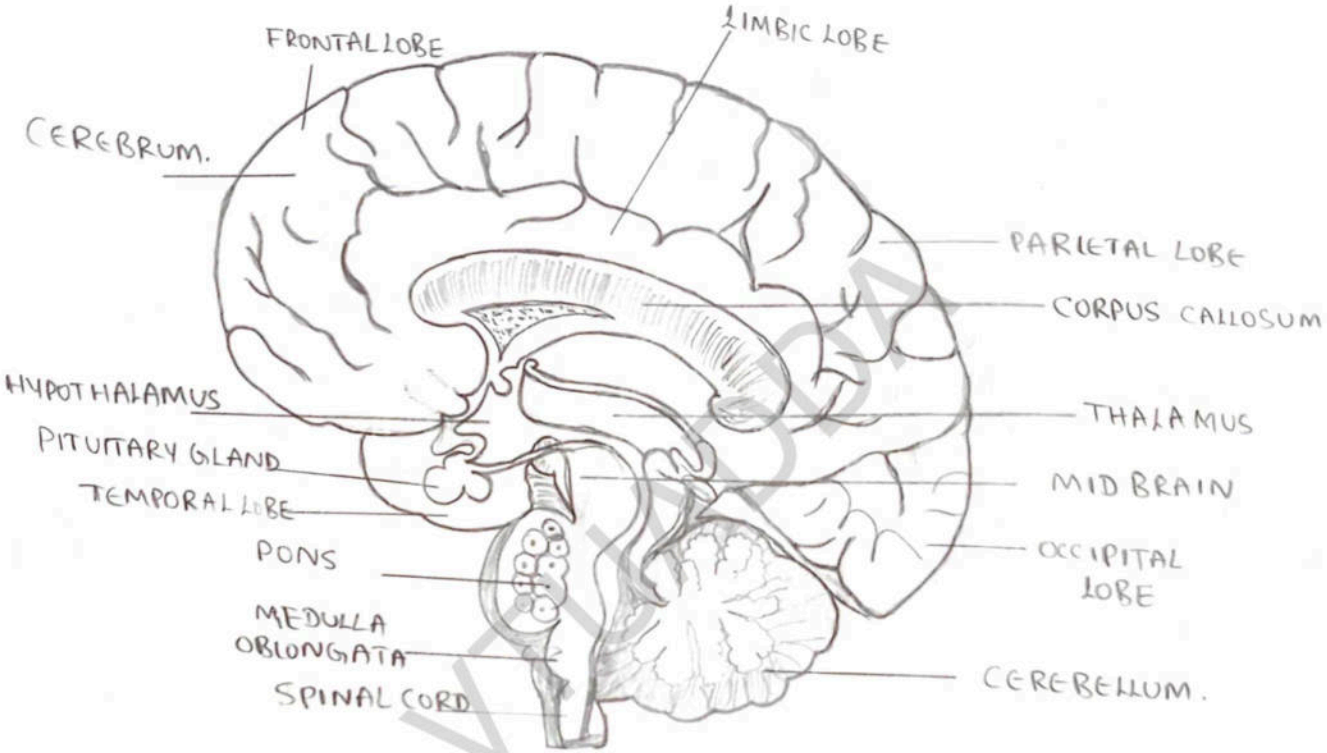

STRUCTURE OF BRAIN

2.1 Brain as a CPU System:

The human brain can be thought of as a highly sophisticated and complex information processing system, similar to a computer's Central Processing Unit (CPU). Both the brain and CPU receive and process inputs, store information, and perform calculations to produce outputs. However, there are significant differences between the two, such as the way they store and process information and the fact that the human brain has the ability to learn and adapt, while a computer's CPU does not. Additionally, the human brain is capable of performing tasks such as perception, thought, and emotion, which are beyond the scope of a computer's CPU.

Table: Comparison Chart

Basis for Comparison	Brain	Computer
Construction	Neurons and synapses	ICs, transistors, diodes, capacitors, transistors, etc.
Memory growth	Increases each time connecting synaptic links	Increases by adding more memory chips
Backup systems	Built-in backup system	Backup system is constructed manually
Memory power		100 million megabytes
Memory density	7	10^{14} bits/cm^3
Energy consumption	12 watts of power	Gigawatts of power
Information storage	Stored in electrochemical and electric impulses.	Stored in numeric and symbolic form (i.e. in binary bits).
Size and weight	The brain's volume is 1500 cm^3 and weight is around 3.3 pounds.	Variable weight and size form few grams to tons.

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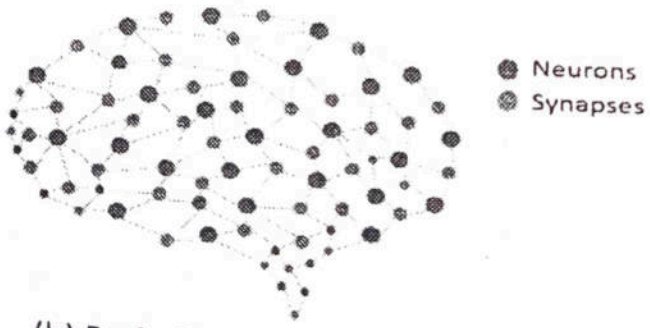
	Uses chemicals to fire the action potential in the neurons.	Communication is achieved through electrical coded signals.
Information processing power	Low	High
Input/output equipment	Sensory organs	Keyboards, mouse, web cameras, etc.
Structural organization	Self-organized	Pre-programmed structure
Parallelism	Massive	Limited
Reliability and damageability properties	Brain is self-organizing, self maintaining and reliable.	Computers perform a monotonous job and can't correct itself.

2.1.1 Architecture

The architecture of the human brain as a CPU system can be compared to that of a parallel distributed processing system, as opposed to the Von Neumann architecture of traditional computers.

(a) Von Neumann Computing System



(b) Brain Computing System

Figure: Comparison between Brains Computing System with Conventional Von Neumann Computing System

2.2 Eye as a Camera System:

Comparison

The human eye can be analogized to a camera system, as both the eye and a camera capture light and convert it into an image.

The main components of the eye that correspond to a camera system include:

- The Cornea: This transparent outer layer of the eye functions like a camera lens, bending light to focus it onto the retina.
- The Iris: The iris functions like the diaphragm in a camera, controlling the amount of light that enters the eye.
- The Pupil: The pupil functions like the aperture in a camera, adjusting the size to control the amount of light entering the eye.
- The Retina: The retina functions like the camera film or sensor, capturing the light and converting it into electrical signals that are sent to the brain.
- The Optic Nerve: The optic nerve functions like the cable connecting the camera to a computer, transmitting the electrical signals from the retina to the brain.

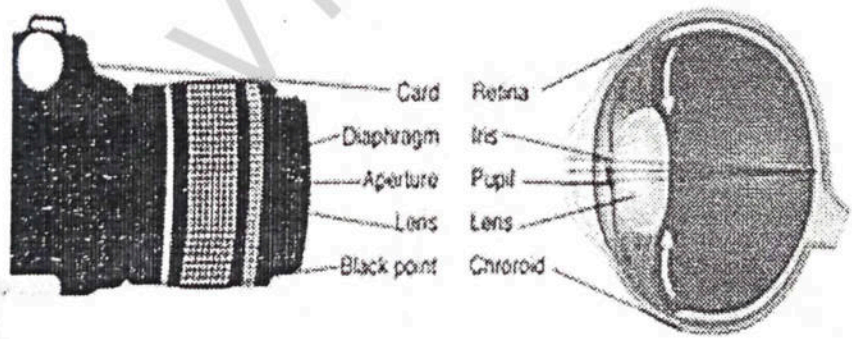


Figure: Comparing camera and anatomy of eye

In both the eye and a camera, the captured light is transformed into an image by the lens and the light-sensitive component. The eye processes the image further, allowing for visual perception, while a camera stores the image for later use.

It's important to note that the eye is much more complex than a camera and has several additional functions, such as adjusting for different levels of light and adjusting focus, that are not found in a

camera. The eye also has the ability to perceive depth and color, as well as adjust to movements and provide a continuous, real-time image to the brain.

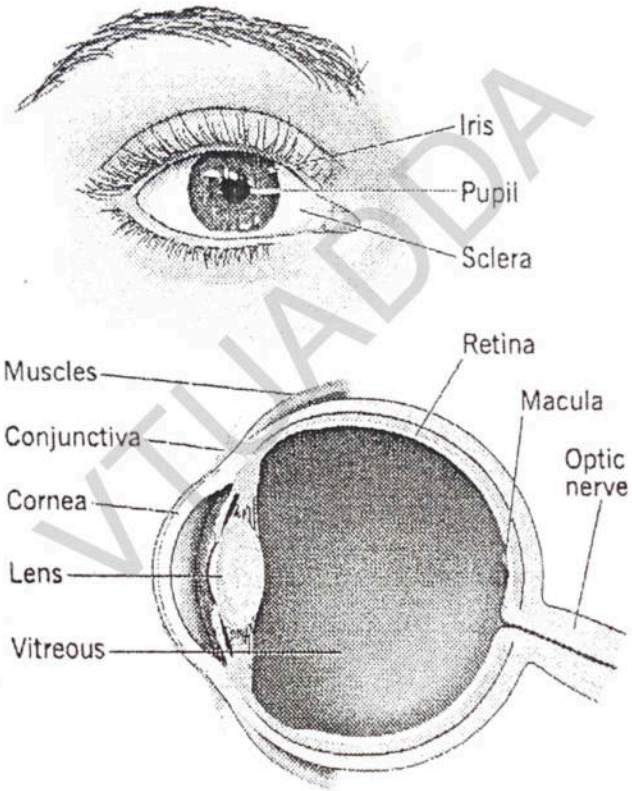
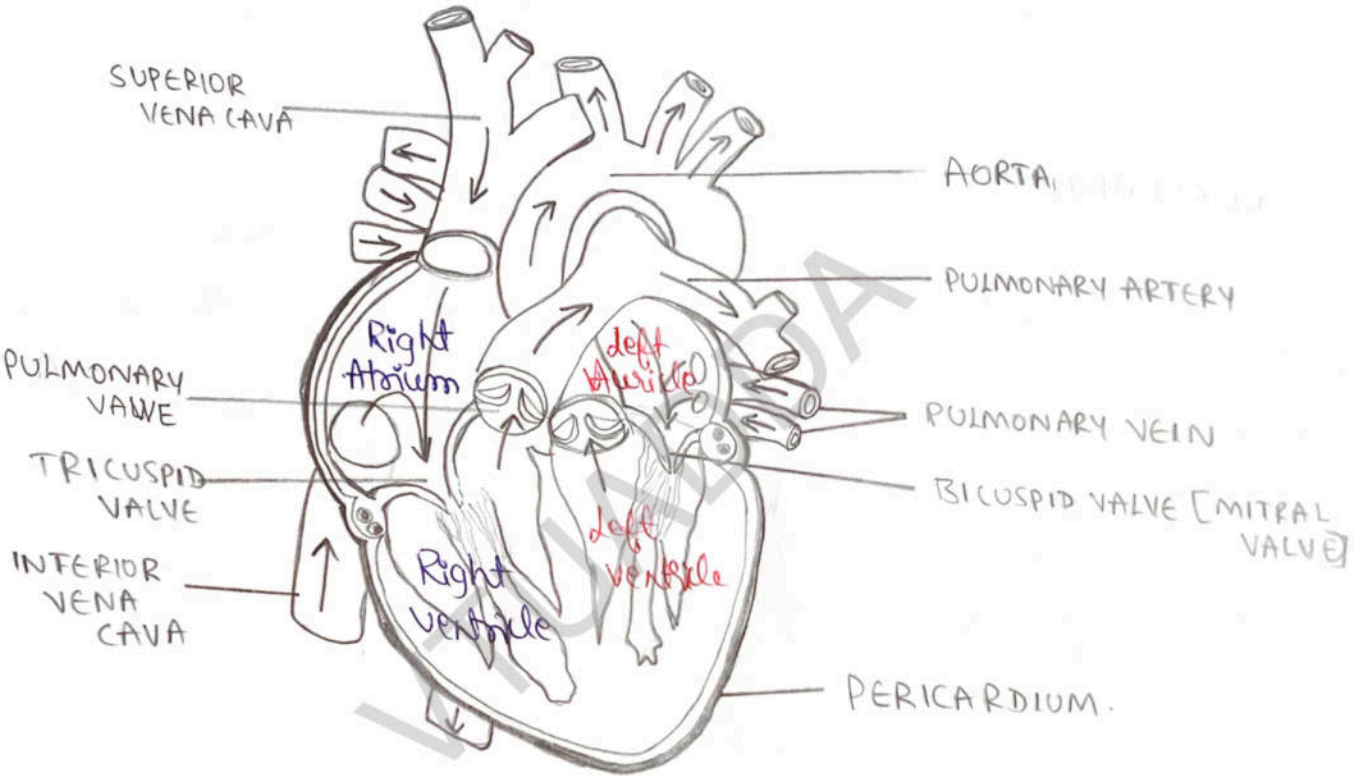


Figure: Representing anatomy of eye



STRUCTURE OF HEART

Architecture of Heart

Heart is surrounded by a sack like structure called pericardium. helps to protect

* Heart is divided into 4 chambers.
* The right atrium, right ventricle, left atrium and left ventricle.
* Right atrium opens right ventricles by tricuspid valve
* Left atrium opens left ventricles via bicuspid valve
* Blood enters the right atrium from the body and is pumped into the right ventricle. Then pumps blood to the lungs for oxygenation.

Oxygenated blood returns to the heart and enters the left atrium. Through pulamonary vein and pumps blood into left ventricle. Left ventricle then pumps the blood out to the rest of the body between each chamber, there are one-way valves that ensure the blood flows in the correct direction and prevent the back flow.

The thick muscle wall running middle of the heart is called septum. It divides auricles and ventricles and also prevent the mixing of oxygenated and deoxygenated blood.

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Semilunar Stops the backflow of blood into heart. the vessels that bring the blood to the heart is Called Veins. The vessels that get rid away blood from heart are Called auricles

4 Blood vessels to take blood in and out of the heart

1) Aorta :- It is largest artery in body it Carries oxygenated blood from the left Ventricle to the body.

2) Vena Cava :- It is a largest vein in the body it Carries deoxygenated blood from the body to heart.

3) Pulmonary artery :- It Carries the deoxygenated blood away from the right ventricle to the lungs.

4) Pulmonary Vein :- Returns oxygenated blood from lungs to heart.

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Architecture of lungs

The lungs is an important Organ in Human Body. It is part of the respiratory System. It is Mainly involved in the process of Breathing.

The primary function of lungs is to facilitate the exchange of Gases between the Air We Breath and Blood Stream.

Trachea - When We Breath, air enters in to the Nose and passes through the Nasal passages to Wind pipe. This Wind pipe is Called Trachea.

It serves as the main passage for air to enter and Exit lungs.

Bronchi - At the entry point of lungs, The trachea divides into two main Branches heading to each lung Called as Bronchi

* The right main Bronchus divides into 3 Smaller Bronchi, Whereas the left main Bronchus divides into two Bronchi These are called Secondary Bronchi which divide further into Tertiary Bronchi.

* Boring Air into Specialised Sections of the lungs.

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(3) Bronchioles - The Bronchi further divide into Small Narrow Branches known as Bronchioles.

Bronchioles Contain Smooth Muscles which can Contract or Expand easily to regulate Airflow.

(4) Alveoli - Alveoli are tiny air Sacs at the end of the Bronchioles. which are the primary sites of Gas Exchange in Lungs.

Each lung contain Millions of Alveoli.

(5) Pulmonary Capillaries - Alveolus are Surrounded by Network of tiny Blood Vesseles Called as pulmonary Capillaries. Allows the Exchange of Gasses between the air in the Alveoli and Blood stream.

(6) Diaphragm - The diaphragm is a dome shaped muscle Located below the Lungs. Plays a Crucial role in Respiration.

(*) It contracts during inhalation this allows the Lungs to expand during inhalation.

(7) Cardiac Notch - The Cardiac Notch is an identation On the Surface of Left Lung. allows the Space for the Heart.

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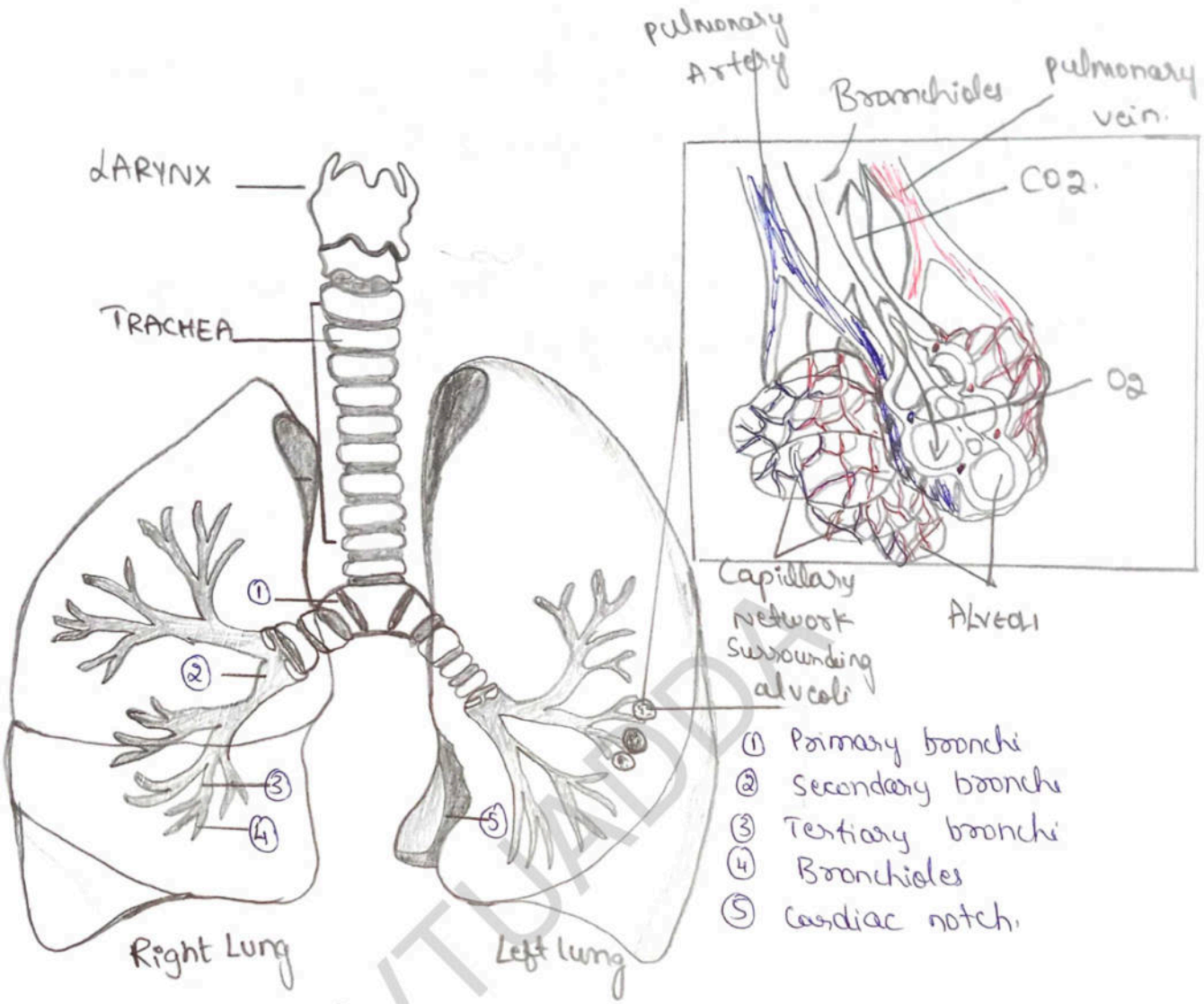
LARYNX
TRACHEA
Right Lung
Left lung
Pulmonary Artery
Bronchioles
pulmonary vein.
CO2.
O2
Capillary network surrounding alveoli
Alveoli
① Primary bronchi
② Secondary bronchi
③ Tertiary bronchi
④ Bronchioles
⑤ Cardiac notch.

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Lungs as Purification System

Every cell in Our body needs Oxygen to live. The air We Breath Contains Oxygen and Other gases. The respiratory Systems main job is to move fresh air into your Body While removing Waste.

① When We Breath, the Nose and Nasal passages act as the first line of defense they trap larger particles such as dust pollen and Pollutants.

* Mucus Lining of the respiratory tract helps to trap and remove particles, Preventing them from reaching deeper into lungs.

* The respiratory System including lungs has an immune defense System that helps to protect against pathogens such as Bacteria, Viruses.

* The Mucus in the respiratory tract contain Anti-bodies and Enzymes that can Neutralise Or destroy pathogens.

* Protecting our body from Harmful substances. This is done by coughing, Sneezing, filtering or Swallo-wing them.

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* Lungs helps in Purification of Blood. Arteries Carry Pure Oxygenated Blood from the Heart to other parts of the Body.

* Veins carry impure Blood, This impure Blood Goes to the Lungs for Purification.

* When the Breath is inhaled Oxygen from the air Comes in Contact With the impure Blood and the Blood Takes up Oxygen.

* The Waste matter in the Blood releases Carbonic acid and The Blood is Purified. The purified blood is Carry to the Heart By Veins.

* Lungs remove Carbon Dioxide and Helps in Maintaining Optimal pH levels of Blood.

Lungs Responsible for

1. Breathing
2. Gas Exchange
3. Oxygenation of Blood
4. Removal of CO_2
5. Defense against foreign particles.

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Gas Exchange Mechanism in lungs

(1). The first step is Contraction of Diaphragm. When Diaphragm Contracts, The muscles in the rib cage Expand the chest Cavity. This Causes Expansion of lungs.

(2). Air enteres into the Nose and Passes through Nasal passage to Trachea.

(3) Then Air passes through Bronchi, Bronchioles and Diffuses across the thin walls of the Alveloi into Blood Capilaries.

(4). When a person inhales lungs inflate and deflate when a person Exhales. During Gas Exchange Oxygen Moves from the lungs to Blood Stream.

At the Same time CO_2 passes from the Blood to the lungs. This Happens in the lungs between the Alveoli and a Network of tiny Blood Vesseles Called capillaries which are located in the Walls of Alveoli.

(*) This lets Oxygen and CO_2 diffuse between the respiratory Stream and Blood stream.

(*) O_2 molecules attaches to Red blood cells travel Back to Heart. At the Same time Carbon dioxide molecules

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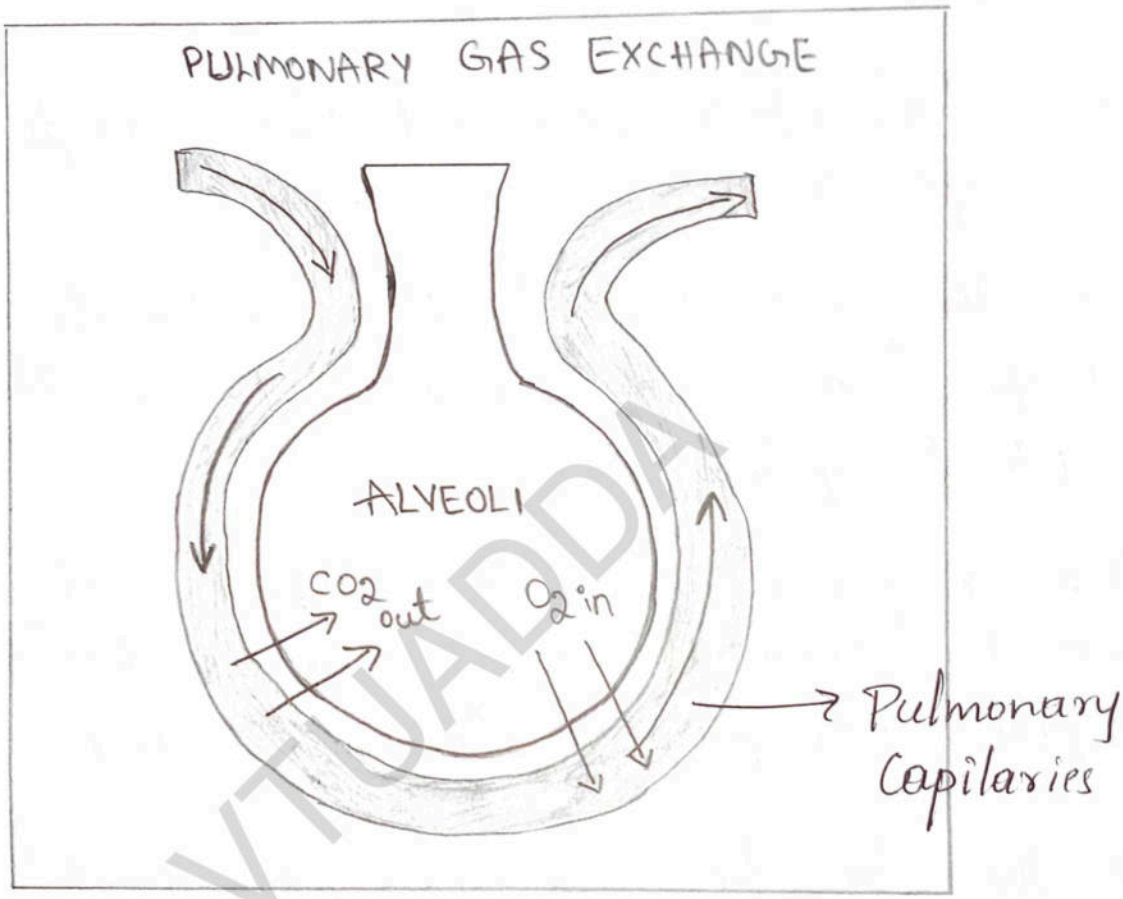
are Blown Out of the Body.



Structure of Alveoli

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Kidney As a Filteration System

Kidneys are two bean-shaped Organs. They are located just below the rib cage, One on each Side of the spine.

The Main role of kidney is to filter out Waste products from Blood stream. Hence it is Considered as a filteration System.

* Architecture of Kidney / Mechanism of Filteration / Kidney as a Filteration System

① Nephron - is the structural and Functional unit of kidney. It is a tube like structure it can be divided in to 2 Major portions.

ⓐ Bowmans Capsule
ⓑ Renal tubule

② Bowman's Capsule - At one end the tube is closed, Expan-ded, folded to a Cup like structure Which is known as Bowmans Capsule.

The Space inside the Bowman's Capsule is known as the Bowman's Space

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Glomerulus - It is a Network of specialised Blood vesseles with small opening in their Walls.

(*) These small pores allows the passage of small Molecules and prevent the passage of larger particles like Blood cells & Protiens. They are Mainly responsible for Filteration of Blood.

(*) The Blood Vessel Which Brings the Blood inside the Glomerulus is known as Afferent Vessel.

(*) The Blood Vessel taking the Blood out of the Glomerulus is Efferent Vessel.

The Bowmans Capsule and Glomerulus together Constitute Renal Corpuscle.

(A) Renal Tubule - A small tube like Structure, long and folded emerges from Bowman's capsule and ends at Duct of the kidney.

It is divided into 3 Main parts Based on its function.

1. Proximal Convoluted Tubule
2. Loop of Henle
3. Distal Convoluted Tubule

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- **Proximal Convoluted Tubule** - Located in the cortex of the kidney. It absorbs Organic Nutrients Glucose & Amino acids and Helps to Maintain Electrolyte levels in Body.

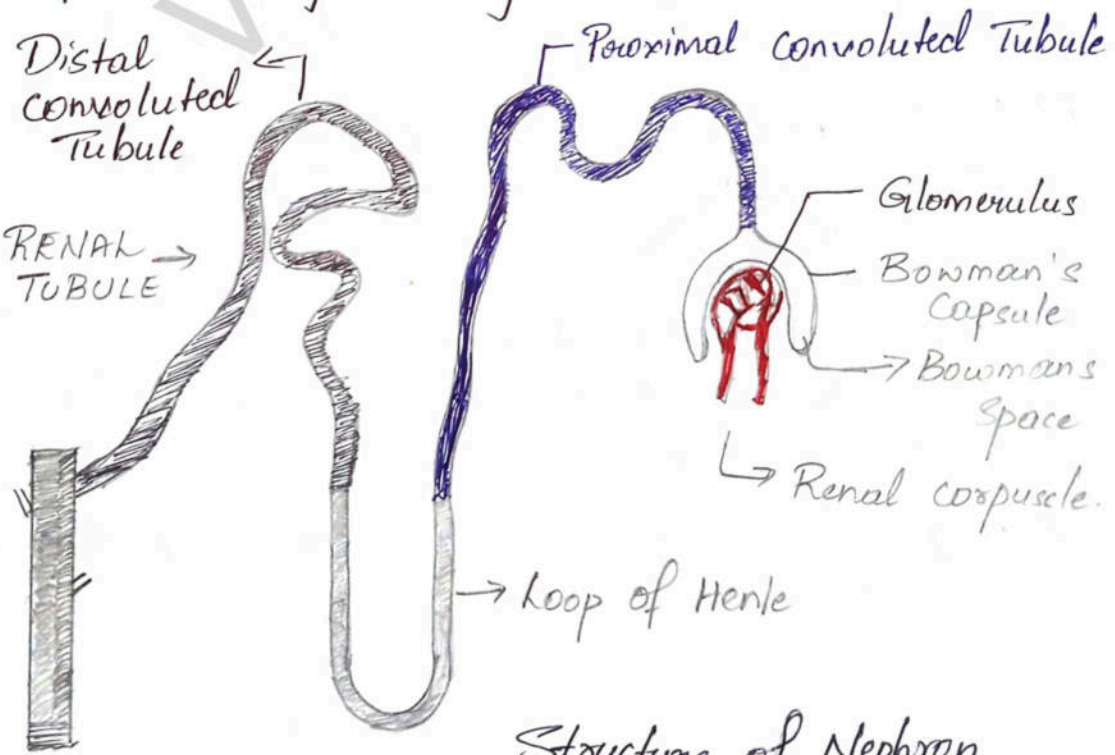
- **Loop of Henle** - Reabsorb water and important Nutrients in the filterate. Located in the Medulla.

- **Distal Convoluted Tubule** - it is also located in the Cortex of the kidney and Helps in Reabsorption.

- **Collecting Duct** - The collecting duct recive Urine from Multiple Nephron.

- **Renal Artery** - The renal Artery is a Blood Vessel through Which impure Blood is Supplied to the Kidney.

- **Renal Vein** - The pure Blood is Supplied to the Other parts of the Body Through Renal Vein.



Structure of Nephron

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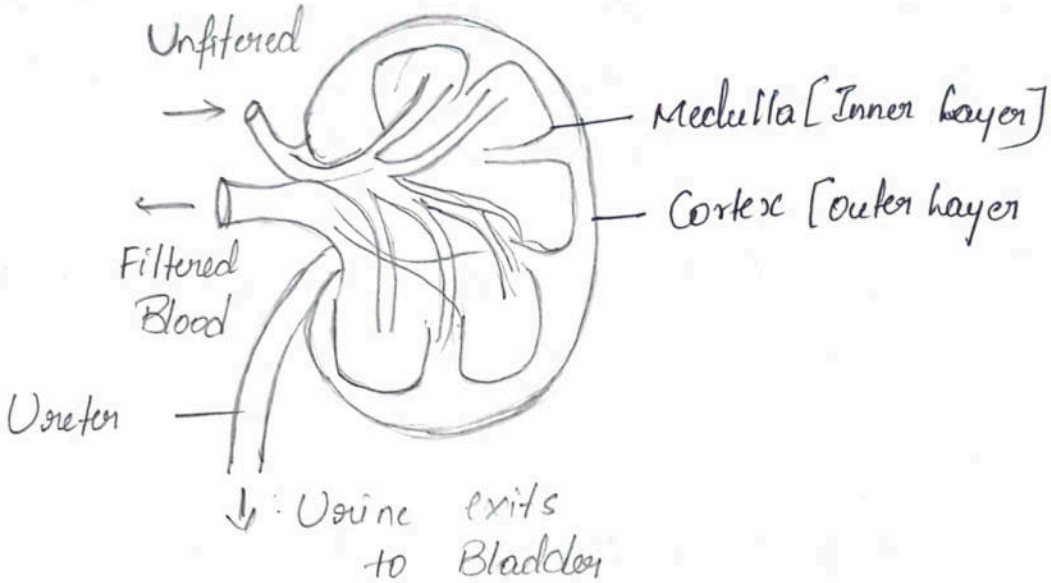

Ureter- Urine is passed from the kidneys to the Urinary Bladder through Two muscular Tubes called Ureters.

Functions

① The primary Function of kidneys is to Filter waste products from Blood stream While Maintaining Electrolytic Balance.

② Blood pressure Regulation- Kidneys produce a Hormone called Renin. that Helps to Control Blood pressure.

③ RBC's Production - Kidneys produce a Hormone called Erythropoietin. which stimulates the Bone Marrow to produce Red Blood cells.

④ Vitamin D Activation- Kidneys play a role in Activating Vitamin D which is important for Calcium Absorption. Helps to Maintain Bone Health.

⑤ kidneys Remove Excess Acid produced by cells Maintain acid Base Balance in the Body.

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